# SEPARATED FINITE-PRESENTATION DIAGONAL OF THE STACK OF POLARIZED PROPER SCHEMES

THE STACKS PROJECT AUTHORS

*Editorial note.* The native definition ofPolarized is included fromquot.tex3765–3800: proper flat finitely presented schemes with a relatively ample invertible sheaf and polarization-preserving isomorphisms.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Proper flat finitely presented families, relatively ample line bundles, representability of Hom/Isom and polarized moduli, cohomology/base change for large tensor powers, projective embeddings and limit criteria for finite presentation.

*Source reference.* Original lemma 0D43: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16438–16438. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0CMW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15871–15871. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original proposition 0CMY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15873–15873. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0DPN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17158–17158. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0DPP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17159–17159. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original section 0DLW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17061–17061. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

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*Date*: stacks-project:moduli.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma 04XD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6391–6391. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 06Q3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8592–8592. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0DNP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17124–17124. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0BEM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14355–14355. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0B9T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14186–14186. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original section 0DPQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17160–17160. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0DPR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17161–17161. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original theorem 0D4X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16467–16467. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original definition 01VH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2446–2446. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original section 04SP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6227–6227. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original section 03YL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5188–5188. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original section 05X6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7610–7610. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

**Lemma 0.1.** *The diagonal of $\mathcal{P}olarized$ is separated and of finite presentation.*

## 1. Properties of the stack of polarized proper schemes

In this section we discuss properties of the moduli stack

$$\mathcal{P}olarized \longrightarrow \mathrm{Spec}(\mathbf{Z})$$

whose category of sections over a scheme $S$ is the category of proper, flat, finitely presented scheme over $S$ endowed with a relatively ample invertible sheaf. This is an algebraic stack by Quot, Theorem 0D4X.

**Situation 1.1.** We define a category $\mathcal{P}olarized$ as follows. Objects are pairs $(X \to S, \mathcal{L})$ where

(1) $X \to S$ is a morphism of schemes which is proper, flat, and of finite presentation, and
(2) $\mathcal{L}$ is an invertible $\mathcal{O}_X$-module which is relatively ample on $X/S$ (Morphisms, Definition 01VH).

A morphism $(X' \to S', \mathcal{L}') \to (X \to S, \mathcal{L})$ between objects is given by a triple $(f, g, \varphi)$ where $f : X' \to X$ and $g : S' \to S$ are morphisms of schemes which fit into a commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{f} & X \\ \downarrow & & \downarrow \\ S' & \xrightarrow{g} & S \end{array}$$

inducing an isomorphism $X' \to S' \times_S X$, in other words, the diagram is cartesian, and $\varphi : f^*\mathcal{L} \to \mathcal{L}'$ is an isomorphism. Composition is defined in the obvious manner (see Examples of Stacks, Sections 04SP and 03YL). The forgetful functor

$$p : \mathcal{P}olarized \longrightarrow \mathcal{S}ch_{fppf}, \quad (X \to S, \mathcal{L}) \longmapsto S$$

is how we view $\mathcal{P}olarized$ as a category over $\mathcal{S}ch_{fppf}$ (see Section 05X6 for notation).

*Proof.* Recall that $\mathcal{P}olarized$ is a limit preserving algebraic stack, see Quot, Lemma 0D43. By Limits of Stacks, Lemma 0CMW this implies that $\Delta : \mathcal{P}olarized \to \mathcal{P}olarized \times \mathcal{P}olarized$ is limit preserving. Hence $\Delta$ is locally of finite presentation by Limits of Stacks, Proposition 0CMY.

Let us prove that $\Delta$ is separated. To see this, it suffices to show that given an affine scheme $U$ and two objects $\upsilon = (Y, \mathcal{N})$ and $\chi = (X, \mathcal{L})$ of $\mathcal{P}olarized$ over $U$, the algebraic space

$$\mathcal{I}som_{\mathcal{P}olarized}(\upsilon, \chi)$$

is separated. The rule which to an isomorphism $\upsilon_T \to \chi_T$ assigns the underlying isomorphism $Y_T \to X_T$ defines a morphism

$$Isom_{\mathcal{P}olarized}(\upsilon, \chi) \longrightarrow Isom_U(Y, X)$$

Since we have seen in Lemmas 0DPN and 0DPP that the target is a separated algebraic space, it suffices to prove that this morphism is separated. Given an isomorphism $f : Y_T \to X_T$ over some scheme $T/U$, then clearly

$$Isom_{\mathcal{P}olarized}(\upsilon, \chi) \times_{Isom_U(Y,X),[f]} T = Isom(\mathcal{N}_T, f^*\mathcal{L}_T)$$

Here $[f] : T \to Isom_U(Y, X)$ indicates the $T$-valued point corresponding to $f$ and $Isom(\mathcal{N}_T, f^*\mathcal{L}_T)$ is the algebraic space discussed in Section 0DLW. Since this algebraic space is affine over $U$, the claim implies $\Delta$ is separated.

To finish the proof we show that $\Delta$ is quasi-compact. Since $\Delta$ is representable by algebraic spaces, it suffice to check the base change of $\Delta$ by a surjective smooth morphism $U \to \mathcal{P}olarized \times \mathcal{P}olarized$ is quasi-compact (see for example Properties of Stacks, Lemma 04XD). We can assume $U = \coprod U_i$ is a disjoint union of affine opens. Since $\mathcal{P}olarized$ is limit preserving (see above), we see that $\mathcal{P}olarized \to \mathrm{Spec}(\mathbf{Z})$ is locally of finite presentation, hence $U_i \to \mathrm{Spec}(\mathbf{Z})$ is locally of finite presentation (Limits of Stacks, Proposition 0CMY and Morphisms of Stacks, Lemmas 06Q3 and 0DNP). In particular, $U_i$ is Noetherian affine. This reduces us to the case discussed in the next paragraph.

In this paragraph, given a Noetherian affine scheme $U$ and two objects $\upsilon = (Y, \mathcal{N})$ and $\chi = (X, \mathcal{L})$ of $\mathcal{P}olarized$ over $U$, we show the algebraic space

$$Isom_{\mathcal{P}olarized}(\upsilon, \chi)$$

is quasi-compact. Since the connected components of $U$ are open and closed we may replace $U$ by these. Thus we may and do assume $U$ is connected. Let $u \in U$ be a point. Let $P$ be the Hilbert polynomial $n \mapsto \chi(Y_u, \mathcal{N}_u^{\otimes n})$, see Varieties, Lemma 0BEM. Since $U$ is connected and since the functions $u \mapsto \chi(Y_u, \mathcal{N}_u^{\otimes n})$ are locally constant (see Derived Categories of Schemes, Lemma 0B9T) we see that we get the same Hilbert polynomial in every point of $U$. Set $\mathcal{M} = \mathrm{pr}_1^*\mathcal{N} \otimes_{\mathcal{O}_{Y \times_U X}} \mathrm{pr}_2^*\mathcal{L}$ on $Y \times_U X$. Given $(f, \varphi) \in Isom_{\mathcal{P}olarized}(\upsilon, \chi)(T)$ for some scheme $T$ over $U$ then for every $t \in T$ we have

$$\chi(Y_t, (\mathrm{id} \times f)^*\mathcal{M}^{\otimes n}) = \chi(Y_t, \mathcal{N}_t^{\otimes n} \otimes_{\mathcal{O}_{Y_t}} f_t^*\mathcal{L}_t^{\otimes n}) = \chi(Y_t, \mathcal{N}_t^{\otimes 2n}) = P(2n)$$

where in the middle equality we use the isomorphism $\varphi : f^*\mathcal{L}_T \to \mathcal{N}_T$. Setting $P'(t) = P(2t)$ we find that the morphism

$$Isom_{\mathcal{P}olarized}(\upsilon, \chi) \longrightarrow Isom_U(Y, X)$$

(see earlier) has image contained in the intersection

$$Isom_U(Y, X) \cap Mor_U^{P', \mathcal{M}}(Y, X)$$

The intersection is an intersection of open subspaces of $Mor_U(Y, X)$ (see Lemma 0DPP and Remark 0DPQ). Now $Mor_U^{P', \mathcal{M}}(Y, X)$ is a Noetherian algebraic space as it is of finite presentation over $U$ by Lemma 0DPR. Thus the intersection is a Noetherian algebraic space too. Since the morphism

$$Isom_{\mathcal{P}olarized}(\upsilon, \chi) \longrightarrow Isom_U(Y, X) \cap Mor_U^{P', \mathcal{M}}(Y, X)$$

is affine (see above) we conclude. □

## References